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1

COMP

Body fluid compartments

You should be able to: State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments.

PICK ONE AND DO THAT ONE

☐ **A** Draw the fluid compartments top down with the fractions. Use one border style for inside cells and a different one for outside.

☐ **B** On a compartment drawing, mark where a blood sample is taken and which compartment you are actually drawing a conclusion about.

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2**Compartment separation and clinical volume shifts**

COMP

You should be able to: Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift.

PICK ONE AND DO THAT ONE

☐ **A** Draw a patient with swollen ankles and a low blood pressure. Use arrows to show where the water went.

☐ **B** Draw three scenarios: water gained in the vessels, water lost from the body, water moved into tissue. Predict blood pressure under each.

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3

Membrane composition and fluidity

COMP

You should be able to: Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

PICK ONE AND DO THAT ONE

☐ **A** Draw a patch of membrane in cross section, big enough to fill half your box. Put in phospholipids with heads and tails clearly different, cholesterol wedged between the tails, a transmembrane protein, a peripheral protein on the inner face, and a glycolipid with its sugar chain. Label which face is extracellular. One line: name the one component whose position alone tells you which side of the membrane you are looking at.

☐ **B** Draw the same membrane patch three times in a row: at normal body temperature, chilled, and warmed. In each, show how tightly the phospholipid tails pack and where cholesterol sits. Draw one tail with a double bond kink in it and one straight saturated tail. One line: cholesterol is called a fluidity buffer and not a stiffener, so say what the word buffer is doing there.

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4

COMP

Determinants of permeability

You should be able to: Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

PICK ONE AND DO THAT ONE

☐ **A** Draw a bilayer as a vertical wall down the middle of your page. Line up six molecules on the left: oxygen, carbon dioxide, water, a steroid hormone, glucose, and a sodium ion. Draw an arrow straight through the wall for each one that crosses unaided, a blocked arrow for each that cannot, and a protein set into the wall for each that needs help. One line: sodium is tiny and still cannot cross, so say what actually stopped it.

☐ **B** Draw a ranking ladder, most permeable at the top and least at the bottom, and place at least six molecules of your own choosing on it. Beside each rung write the properties that put it there: size, lipid solubility, and charge where it applies. One line: name the pair on your ladder that are close in size but far apart in rank, and say what separates them.

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5

COMP

Simple diffusion and Fick's law

You should be able to: State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer.

PICK ONE AND DO THAT ONE

☐ **A** Draw the Fick's law setup as a picture, not an equation: two compartments, a membrane between them, and label on the drawing the surface area, the concentration difference, and the membrane thickness. Then draw four small graphs in a row, rate of diffusion against each of those, plus against diffusion distance. Label both axes on every graph. One line: one variable sits in the denominator, so say which and how the shape of its graph shows that.

☐ **B** Draw a real gas exchange surface, one alveolus against one capillary, and mark on it every Fick variable the body has already optimised. Then draw the same surface in pulmonary edema, with fluid thickening the barrier. Show what changed and what did not. One line: name the single Fick variable the disease changed, and say whether breathing faster fixes it.

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6

COMP

Osmolarity and tonicity

You should be able to: Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell.

PICK ONE AND DO THAT ONE

☐ **A** Draw three beakers labeled 150 mM NaCl, 300 mM sucrose, and 300 mM urea. Under each, write the osmolarity calculation and the answer in mOsm/L. Then draw a red blood cell in each beaker and show its final volume. One line: all three have the same osmolarity and one of them lyses the cell, so name the property of the solute that decides tonicity.

☐ **B** Draw three cells that have just come out of three unknown solutions: one shrunken, one swollen but intact, and one lysed. Beside each, write two different solutions that could have produced that result, one differing in osmolarity and one differing only in penetrating solute content. One line: say what osmolarity alone failed to tell you about the lysed cell.

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7

COMP

Osmosis and cell volume

You should be able to: Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content.

PICK ONE AND DO THAT ONE

☐ **A** Draw a cell at normal volume in the middle of your page, 290 mOsm/L inside. Around it draw four more cells, in 150 mOsm/L NaCl, 290 mOsm/L NaCl, 450 mOsm/L NaCl, and 290 mOsm/L urea. For each, draw the water arrow with its direction and the final cell volume. One line: for the urea cell, say what happens at the first instant and what happens over the next few minutes.

☐ **B** Draw one set of axes, cell volume on the y and time on the x, with the moment of immersion marked. Plot three traces on it: a cell in 450 mOsm/L NaCl, a cell in 150 mOsm/L NaCl, and a cell in 600 mOsm/L urea. One line: one trace goes down and then climbs back past where it started, so say which solute movement caused each half of that curve.

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8

Diffusion and osmosis experiment

COMP

You should be able to: Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.

PICK ONE AND DO THAT ONE

☐ **A** Before you run anything, draw your predicted result: the bag or the plate as it will look at the start and at the end, with each test molecule drawn where you expect it to be. Run it, then draw the actual result beside your prediction in a second color. One line: name the substance that moved fastest and say whether molecular size or starting gradient explains it better in your own data.

☐ **B** Draw a graph of distance moved on the y against molecular weight on the x, one labeled point for every substance you tested. Beside it draw a second small graph, distance moved against starting concentration. One line: your membrane passed some things and stopped others, so write the size cutoff your data actually supports and how confident you are in that number.

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9

COMP

Tonicity and red blood cells

You should be able to: Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.

PICK ONE AND DO THAT ONE

☐ A Draw three microscope fields, one per solution: distilled water, 0.9 percent NaCl, and 3 percent NaCl. In each, draw the cells as you actually see them, keeping the biconcave disc, the spiky crenated form, and the ghosts distinct. Under each field write the direction of net water movement. One line: name the slide that is hardest to count cells on, and say why that difficulty is itself the result.

☐ B Draw one red cell in the center of your box and three arrows out to three versions of it: crenated, normal, and hemolysed. Beside each write the tonicity that produced it and the water arrow. Then draw what the membrane itself did in the hemolysed one. One line: crenation is reversible and hemolysis is not, so say what makes the difference.

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10

COMP

Facilitated diffusion

You should be able to: Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.

PICK ONE AND DO THAT ONE

☐ **A** Draw a channel protein and a carrier protein in the same stretch of membrane, with a molecule crossing through each. Make the difference visible in the drawing: the channel as an open pore, the carrier changing shape around its cargo. Underneath, draw one graph of transport rate against solute concentration with two traces, simple diffusion and carrier mediated. One line: only one trace plateaus, so say what has run out at the plateau.

☐ **B** Draw a carrier protein with glucose bound, then draw three failures beside it: galactose competing for the same carrier, a molecule of the wrong shape bouncing off, and every carrier already occupied. Label which drawing is competition, which is specificity, and which is saturation. One line: adding more glucose helps in one of those three and does nothing in another, so name each and say why.

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11**Primary active transport**

COMP

You should be able to: Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.

PICK ONE AND DO THAT ONE

☐ **A** Draw the sodium potassium ATPase in a membrane with inside and outside labeled. Sketch the cycle in three or four steps: ions binding, ATP splitting, the shape change, ions releasing. Write 3 and 2 on the correct arrows in the correct directions. Beside it note the sodium and potassium concentrations inside and out that the pump maintains. One line: it moves three out for every two in, so say what that does to charge across the membrane.

☐ **B** Draw a cell with the two gradients the pump maintains, sodium high outside and potassium high inside, as tall and short bars on each side. Now poison the pump with ouabain and draw the same bars ten minutes later and an hour later. One line: pick one downstream process that fails, not the resting membrane potential, and say how many steps removed from the pump it is.

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12

COMP

Secondary active transport

You should be able to: Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.

PICK ONE AND DO THAT ONE

☐ **A** Draw a cell membrane with the sodium potassium ATPase keeping sodium low inside. Beside it draw SGLT carrying sodium and glucose in together, and the sodium calcium exchanger sending calcium out as sodium comes in. Draw every ion arrow, and mark the sodium gradient as the shared energy source. Label which one is symport and which is antiport. One line: no ATP touches either of these two proteins, so say where their energy came from.

☐ **B** Draw the sodium calcium exchanger working normally, then draw the same exchanger in a cell whose sodium potassium pump has been stopped. Use three panels with arrows between them: what happens to intracellular sodium, then to the exchanger's driving force, then to intracellular calcium. One line: this is how digoxin strengthens a failing heart, so write the last step of that chain in your own words.

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13**Transport maximum and saturation**

COMP

You should be able to: Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.

PICK ONE AND DO THAT ONE

☐ **A** Draw one set of axes, transport rate on the y and solute concentration on the x. Draw a carrier mediated curve that rises and then flattens, and mark the transport maximum with a labeled dashed line. On the same axes draw a straight simple diffusion line for contrast. One line: past the transport maximum the curve is flat, so say what is happening to every extra solute molecule that arrives.

☐ **B** Draw a kidney tubule with glucose filtered in, reabsorbed by carriers, and any leftover carried on to urine. Draw it three times: normal plasma glucose, plasma glucose at the renal threshold near 180 mg/dL, and plasma glucose well above that. Show the urine in each. Write the glucose transport maximum, about 375 mg/min, beside them. One line: say why a person can be losing glucose in urine while every carrier is working perfectly.

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14**Vesicular transport**

COMP

You should be able to: Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.

PICK ONE AND DO THAT ONE

☐ **A** Draw one cell with four events happening on its surface at once: phagocytosis of a bacterium, pinocytosis of a droplet of fluid, receptor mediated endocytosis of a particle held in a coated pit, and exocytosis of a secretory vesicle. Draw the membrane actually deforming in each. Beside each write the trigger and the cargo. One line: all four cost ATP, so name the one that is completely unselective about what it takes in, and say what that costs the cell.

☐ **B** Draw one particle's whole journey by receptor mediated endocytosis, left to right across your page: binding to receptors, the coated pit, the vesicle pinching off, fusion with a lysosome, and the receptors returning to the surface. Underneath, draw the reverse journey for a molecule leaving by exocytosis. One line: pick a cell type, not a neuron, and say which direction it spends more energy on.

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15**Transepithelial transport**

COMP

You should be able to: Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.

PICK ONE AND DO THAT ONE

☐ **A** Draw one polarized epithelial cell, lumen on the left and blood on the right, tight junctions at the top, apical and basolateral membranes labeled. Trace one glucose molecule all the way through: SGLT at the apical membrane, GLUT at the basolateral membrane, and the sodium potassium pump on the basolateral side. Mark each step active or passive. One line: the pump never touches glucose, so say how it powers the apical step.

☐ **B** Draw the same epithelium twice. In the first, show what would go wrong if the basolateral pump drifted around to the apical side, and use that to say what tight junctions are for. In the second, draw the cell with apical SGLT blocked. Show where glucose ends up in each. One line: oral rehydration solution contains both salt and sugar, so say which of your two drawings explains why.

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16

COMP

Transport simulation

You should be able to: Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.

PICK ONE AND DO THAT ONE

☐ **A** Before you run it, draw three predicted graphs in a row, transport rate against time, one each for simple diffusion, facilitated diffusion, and active transport. Then draw a fourth showing what a metabolic poison does to whichever of the three it affects. Run the simulation and draw the real curves underneath in a second color. One line: say which transport type the poison left untouched, and why it could not touch it.

☐ **B** Draw a decision tree with three endpoints: simple diffusion, facilitated diffusion, and active transport. At each branch put one test you can actually run in the simulation, reversing the gradient, raising solute until the rate stops climbing, or adding a metabolic poison. Draw the result that sends you down each branch. One line: name the single test that separates active transport from both of the others in one step.

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17**Signal types and range**

COMP

You should be able to: Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

PICK ONE AND DO THAT ONE

☐ **A** Draw one target cell in the middle of the box. Draw an arrow in from each of the five signal types, autocrine, paracrine, neurotransmitter, neurohormone, hormone. Label each arrow with how far it traveled and what carried it.

☐ **B** Make a five row table. Signal type, where it is released, how far it goes, what it travels through. Fill it from memory, then check one row against the book.

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18**Receptor location and ligand solubility**

COMP

You should be able to: Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

PICK ONE AND DO THAT ONE

☐ **A** Draw a cell with one surface receptor and one intracellular receptor. Show the path a lipid soluble signal takes and the path a water soluble one takes. Mark which response is faster and which lasts longer.

☐ **B** Write the two questions you would ask about any new signal molecule to predict where its receptor sits. Then answer both for a steroid and for a peptide hormone.

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19

COMP

G protein coupled receptors

You should be able to: Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

PICK ONE AND DO THAT ONE

☐ **A** Draw the membrane with a receptor crossing it, and trace the pathway in numbered order: ligand binds, the G protein swaps GDP for GTP, the alpha subunit leaves, it switches on the amplifier enzyme, the enzyme makes second messenger, the messenger switches on a kinase, the kinase changes a target protein. Beside each step write roughly how many molecules that one step produced from one. One line: name the step where the biggest multiplication happened.

☐ **B** Draw a Gs pathway and a Gq pathway side by side, starting from the same membrane and ending at two different cellular responses. Label the amplifier enzyme and the second messenger on each. Then draw the off switch on both, GTP being hydrolyzed on the alpha subunit. One line: cholera toxin jams that off switch open, so write what the gut cell keeps doing and why the person loses fluid.

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20

COMP

Second messengers

You should be able to: Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

PICK ONE AND DO THAT ONE

☐ **A** Draw a membrane and put all four second messengers on one page: cyclic AMP, IP3, diacylglycerol, and calcium. For each, draw where it comes from, the enzyme or channel that produced it, and an arrow to one thing it switches on. Keep DAG in the membrane and IP3 out in the cytosol, because that is where they actually go. One line: one of the four is not made by an enzyme at all, so name it and say where it comes from instead.

☐ **B** Draw phospholipase C cutting PIP2 in the membrane and follow both halves. Send IP3 to the endoplasmic reticulum and draw what comes out. Leave DAG where it is and draw what it activates. Show the two halves converging on one response. Draw the adenylyl cyclase and cyclic AMP route beside it for contrast. One line: cyclic AMP is destroyed by phosphodiesterase, so say what a drug that blocks that enzyme does to the signal.

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21

Catalytic receptors and intracellular receptors

COMP

You should be able to: Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

PICK ONE AND DO THAT ONE

☐ **A** Draw one cell with the insulin receptor in the membrane on one side and a steroid receptor sitting inside the cell on the other. Trace both from hormone arrival to final response, the steroid crossing the membrane and reaching DNA, the insulin receptor phosphorylating itself and moving a glucose transporter to the surface. Put a clock beside each showing how long the response takes. One line: name the one you could not use if you needed the effect ten seconds from now.

☐ **B** Draw one time axis with two response curves on it, a receptor enzyme response and a steroid response, each from hormone arrival through peak to fade. Make the onsets and durations honestly different. Under the axis, draw what is physically being changed in each case, an existing protein being modified or a new protein being built. One line: say which of the two you would want for the next thirty seconds and which for the next week.

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22

COMP

Signal amplification

You should be able to: Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

PICK ONE AND DO THAT ONE

☐ **A** Draw a three step cascade as a widening funnel. Put a number on each step and multiply them down the side to get the total amplification.

☐ **B** Start from one ligand and write the count at each step of a cascade until you reach the response. One line per step, numbers only, no sentences.

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23

COMP

Receptor modulation

You should be able to: Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

PICK ONE AND DO THAT ONE

☐ **A** Draw one target cell three times. Normal, after a long stretch of scarce signal, after a long stretch of excess signal. Show the receptor count changing in each and label which is up regulation and which is down.

☐ **B** Define agonist, antagonist, up regulation and down regulation in your own words, one line each. Then explain in two lines why a patient who needs more opioid for the same effect is usually showing down regulation rather than drug seeking.

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24

COMP

Signal termination

You should be able to: Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

PICK ONE AND DO THAT ONE

☐ **A** Draw one signal running from receptor to response across a cell, then add three different off switches to that same drawing at three different points: the ligand removed or destroyed outside the cell, the second messenger broken down inside, and the response protein having its phosphate taken off. Name the enzyme doing each. One line: pick the switch nearest the response and say what still goes wrong if only that one fails.

☐ **B** Pick a signal, not insulin, and draw it working normally, then draw it again with its termination broken. In the broken version show what piles up and where. Underneath, draw the effect on the tissue as two panels, one over minutes and one over weeks. One line: a signal that never shuts off eventually produces less response rather than more, so write what the cell did to itself.

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Dose response relationships

You should be able to: Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

COMP

PICK ONE AND DO THAT ONE

☐ A Draw a dose response curve with log dose on the x axis and response as a percentage of maximum on the y. Mark threshold, EC50, and the maximal response, with construction lines dropped to the axes. Then plot a partial agonist on the same axes. One line: the partial agonist never reaches the top no matter how much you add, so say what that tells you about the drug that raising the dose cannot fix.

☐ B Draw three curves on one set of axes: a full agonist alone, the same agonist with a competitive antagonist added, and the same agonist with an antagonist that will not come off the receptor. Show which curve slides sideways and which one gets shorter. One line: only one of the two antagonists can be beaten by giving more agonist, so name it and say what feature of the graph proves it.

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